

Boini Abhiram

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SUMMARY

Data Science & AI undergraduate with production ML experience — a deployed government AI system and three live, end-to-end projects spanning MLOps, causal inference, and agentic LLM systems. Comfortable owning models in production: serving, monitoring, drift-triggered retraining, fine-tuning (QLoRA), and rigorous evaluation. Seeking remote ML / Data Science / AI engineering work.

EDUCATION

International Institute of Information Technology, Naya Raipur (IIIT-NR)

B.Tech in Data Science and Artificial Intelligence — CGPA: 8.70/10.0

Naya Raipur, India

Expected Apr 2028

EXPERIENCE

Research Intern

May 2026 – July 2026

Indian Institute of Technology (BHU), Varanasi

Varanasi, India

- Researching urban morphology in India's Tier-II cities via multi-modal semantic segmentation of high-resolution satellite imagery.
- Building deep-learning segmentation models to classify urban land-use, building footprints, road networks, and green cover across Indian cities.
- Applying multi-modal fusion of optical and SAR satellite data to improve segmentation accuracy in heterogeneous urban environments.

AI/ML Engineer

Dec 2025 – March 2026

Department of Forests, Chhattisgarh Government

Raipur, India

- Built and deployed to production a Hindi voice-based animal-health logging system for a government zoo, enabling non-literate zookeepers to record observations by speech.
- Integrated Deepgram ASR for speech-to-text and Gemini LLM for automatic summarization and structured record generation from voice inputs.
- Engineered the end-to-end pipeline (audio ingestion → database) with role-based modules for inventory, medication tracking, and veterinary workflows.

AI Data Analyst (Remote)

Dec 2025 – Jan 2026

Rochester Institute of Technology

New York, USA

- Modeled student-engagement and drop-off behavior on an imbalanced retention dataset; handled class imbalance with class weighting and reported precision/recall and PR-AUC to surface the top-risk cohort for intervention.
- Built interactive Tableau dashboards translating model output into retention actions for program stakeholders.

AI/ML Engineer Intern (Remote)

Oct 2025 – Dec 2025

Infosys Springboard

Remote

- Built SmileAgeAI, a privacy-first real-time facial-attribute app (smile classification + age regression) over webcam and image uploads.
- Designed custom CNNs (smile F1 0.87; age MAE 5.55 yrs) and an optimized inference pipeline using Haar Cascade detection and CLAHE enhancement.

PROJECTS

Real-Time MLOps Fraud Detection & Drift Monitoring [Live Demo]

MLflow, FastAPI, Docker, CI/CD

- Built an end-to-end production system that serves real-time fraud predictions at **test PR-AUC 0.77** (precision 0.87 / recall 0.75) on a ~0.17%-positive dataset, handling imbalance with `scale_pos_weight` rather than naive resampling.
- Closed the MLOps loop: Evidently drift detection triggers retraining, an MLflow-registry promotion gate promotes the challenger *only if it beats* the incumbent on the same current data, and the FastAPI service hot-reloads the new version.
- Containerized the full stack (API + MLflow + monitoring) to run from a clean clone with `docker compose up`; added ruff + pytest CI via GitHub Actions.

Multi-Agent Financial Analyst with Evals & Fine-Tuned SQL Tool [Live Demo] *LangGraph, QLoRA, RAGAS*

- Built an autonomous LangGraph multi-agent system (planner → researcher → analyst) that plans a task, routes across tools (10-K retrieval, web search, Text-to-SQL), and produces a cited analytical report from real SEC EDGAR data.

- Fine-tuned a Text-to-SQL model with **QLoRA** (4-bit, r=16) on Spider, lifting execution accuracy from 49.0% to 54.5% (+11% rel.) on 200 held-out dev queries, and integrated it as a live agent tool.
- Added input/output guardrails (prompt-injection, PII, read-only SQL), a human-in-the-loop checkpoint, and a RAGAS + DeepEval evaluation harness with LangSmith tracing for every run.

Causal Inference & A/B Experimentation Platform [\[Live Demo\]](#) *DoWhy, EconML, statsmodels, Streamlit*

- Built a self-serve platform that runs A/B design checks (power, sample-ratio mismatch), frequentist analysis with CUPED variance reduction, and causal methods (difference-in-differences, propensity-score matching, causal forests).
- Demonstrated bias correction: on confounded data the naive A/B estimate overstated the true effect 5× (11.0 vs 2.0) while propensity-score matching recovered it (1.6); every method is unit-tested against a known simulated ground-truth effect.
- Output a plain-English ship / don't-ship verdict combining significance, effect size, and sample adequacy.

RESEARCH

Multimodal Stress Detection using Deep Learning | *IEEE paper, in progress*

- Built a non-intrusive multimodal stress-detection system fusing facial-visual and acoustic-speech cues; the hybrid model (3D-CNN + EfficientNet + Action-Unit gating) reached 94.7% accuracy, vs. 87.3% unimodal baseline.
- Designed the preprocessing pipeline (MediaPipe face alignment, voice-activity detection, augmentation), reducing error rate 38% over the baseline.

TECHNICAL SKILLS

Languages: Python, SQL, R, C++, Java

ML / DL: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, CNNs, fine-tuning (LoRA/QLoRA, PEFT)

LLM / AI: LangGraph, RAG, RAGAS, DeepEval, LangSmith, vector DBs (Chroma), prompt engineering, embeddings

MLOps / Serving: MLflow, FastAPI, Docker, GitHub Actions (CI/CD), Evidently AI, model monitoring

Data Science: causal inference (DoWhy, EconML), A/B testing, statsmodels, EDA, feature engineering, time-series

Tools: Git/GitHub, Linux, Streamlit, Gradio, Tableau, Hugging Face

CERTIFICATIONS & ACHIEVEMENTS

Kaggle Expert (Notebooks & Datasets) | IBM Data Science Professional Certificate | Oracle Cloud Infrastructure 2025 Data Science Professional | Google Gen AI Academy 2.0 | JEE Main: 97.7 percentile